Today

Polynomial Regression

General Linear Least-squares

Test error vs. Training Error

Announcements

- Exam 3 next Wednesday
  - covers through today (with possible minor additions Monday)
  - Review materials released tomorrow
- Lab 4 due tomorrow
  - No more assignments due until after Thanksgiving
- Lab 5: teams assigned randomly
  - Will turn in first part in class

Polynomial Regression

Recall Linear Regression: $\hat{y} = m_{lin}(x;\theta) = a_0 + a_1 x$

$\theta = (a_0, a_1)$

Poly. Reg.: $\hat{y} = m_{poly}(x;\theta) = a_0 \underbrace{x^0}_{1} + a_1 x + a_2 x^2 \dots a_m x^m$

$\theta = \vec{a} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix}$

use $l(\hat{y}_i - y_i) = (\hat{y}_i - y_i)^2$

minimize $J(\theta; X, Y) \Rightarrow$ minimize $\frac{1}{n}\sum_{i=1}^{n}(a_0 + a_1 x_i + \dots a_m x_i^m - y_i)^2$

θ $\quad$ $\vec{a}$

F.O.N.C. $\nabla J(\theta) = \vec{0} = \begin{bmatrix} \frac{\partial J}{\partial a_0} \\ \vdots \\ \frac{\partial J}{\partial a_m} \end{bmatrix}$

$z_{0i} = 1 \quad z_{1i} = x_i \quad z_{2i} = x_i^2 \quad \dots \quad z_{ji} = x_i^j \quad \dots \quad z_{mi} = x_i^m$

$J(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sum_{j=0}^{m} a_j z_{ji}\right)^2$

$\frac{\partial J}{\partial a_j} = \frac{2}{n}\sum_{i=1}^{n}\left(y_i - \sum_{j=0}^{m} a_j z_{ji}\right) z_{ji} = 0$

$Z = \begin{bmatrix} z_{01} & z_{11} & \dots & z_{m1} \\ z_{02} & & & \\ \vdots & & & \\ z_{0n} & \dots & & z_{mn} \end{bmatrix}^{T} = \begin{bmatrix} 1 & x_1 & \dots & x_1^m \\ & & & \\ & & & \\ 1 & x_n & \dots & x_n^m \end{bmatrix}$ ← first row: x_1

can rearrange to

$Z^T Z \vec{a}^* - Z^T Y = \vec{0}$ ($n \times 1$ matrix)

$Z^T Z$ computed from X

only unknowns are $\vec{a}$, solve for $\vec{a}$

$$\vec{a}^* = (Z^T Z) \backslash Z^T Y$$

Let $m = 1$

$Z = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \\ 1 & x_n \end{bmatrix}$ $\quad$ $Z^T Z = \begin{bmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_n \end{bmatrix}\begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{bmatrix}$

Solved F.O.N.C. to get $\theta^* = \vec{a}^*$

What about second order conditions

$H = 2Z^T Z$

can be shown if Z has full column rank $Z^T Z$ is pos def

H is pos def $\Rightarrow$ second order sufficient condition ✓

$\Rightarrow$ Local min

can be shown that $J(\theta)$ is convex

$\Rightarrow$ Local min is global min.

How do we choose what order m to use?

Handout

$$RMSE_{train} = \sqrt{\frac{1}{n}\sum_{i \in train}(\hat{y}_i - y_i)^2} = \sqrt{J(\theta)}$$

train : points used to fit curve

test : points that you want to predict for

$$RMSE_{test} = \sqrt{\frac{1}{n}\sum_{i \in test}(\hat{y}_i - y_i)^2}$$ follows $\cup$ shape as m changes

Underfit: Both train error and test error are high — m too low

Good fit: test error is near minimum and only slightly higher than train error — m just right

Overfit: ~~training~~ test error is much higher than ~~test~~ training error — m too high